Week 2. ER Modeling

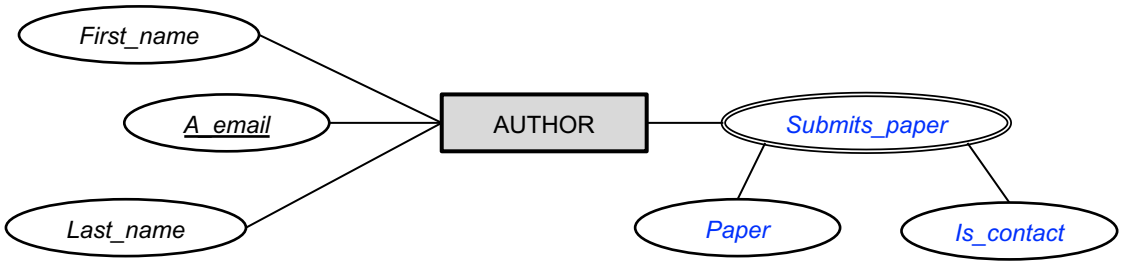
Exercise 2: ER Schema Diagram (Cont.)

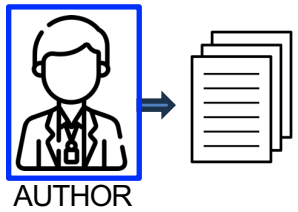
❑ Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

❑ Entity Type 1: AUTHOR

- Requirements:
  - Authors of papers are uniquely identified by e-mail id and have the first and last names
  - Authors can submit one or more papers
    - (Paper) Each can have multiple authors, but one of the authors is designated as the contact author
- Attributes:
  - *A_email, First_name, Last_name, {Submits_paper(Paper, Is_Contact)}*





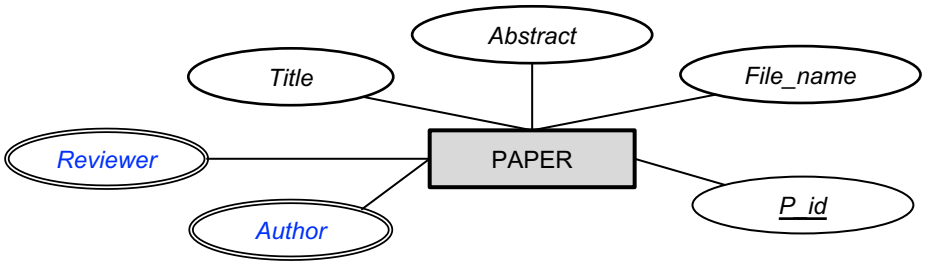
Exercise 2: ER Schema Diagram (Cont.)

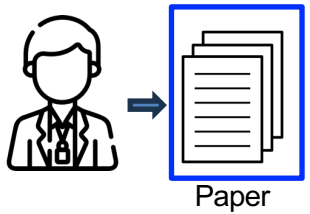
❑ Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

❑ Entity Type 2: PAPER

- Requirements:
  - Each paper is assigned a unique id by the system and described by a title, abstract, and the file name
  - Each can have multiple authors, but one of the authors is designated as the contact author
  - A paper is assigned between two and four reviewers
- Attributes:
  - *P_id, Title, Abstract, File_name , {Author}, {Reviewer}*





Exercise 2: ER Schema Diagram (Cont.)

Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

Entity Type 3: REVIEWER

- Requirements:
  - Reviewers of papers are uniquely identified by e-mail id
  - Each reviewer has the first and last name, phone number, affiliation, and topics of interest
  - A reviewer rates each paper assigned to her/him on a scale of 1 to 10 in four evaluation categories: (1) technical merit, (2) readability, (3) originality, and (4) relevance to the conference.
  - Each reviewer provides an overall recommendation to each paper assigned to her/him
  - Each review contains two types of written comments: (1) one to be seen by the review committee only and (2) the other as feedback to author(s)
- Attributes:
  - *R_email, First_name, Last_name, Phone_no, Affiliation, {Interest_topic}, {Reviews_paper(Paper, Recommendation, Comments(Feedback, For_committee), Rating(C1:Technical_merit, C2:Readability, C3:Originality, C4:Relevance))}*

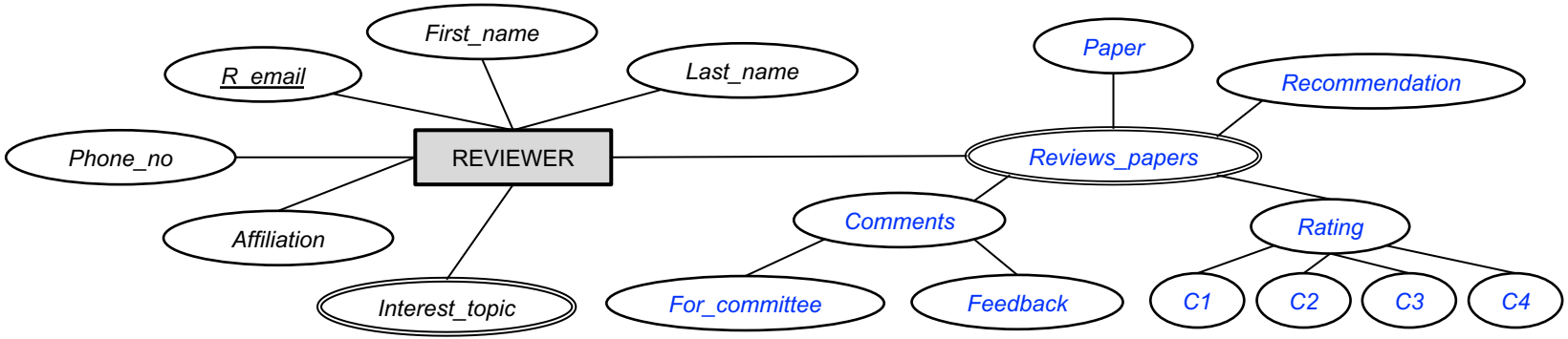
Exercise 2: ER Schema Diagram (Cont.)

Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

Entity Type 3: REVIEWER

- Attributes:
  - *R_email*, *First_name*, *Last_name*, *Phone_no*, *Affiliation*, *{Interest_topic}*, *{Reviews_paper(Paper, Recommendation, Comments(Feedback, For_committee), Rating(C1:Technical_merit, C2:Readability, C3:Originality, C4:Relevance))}*



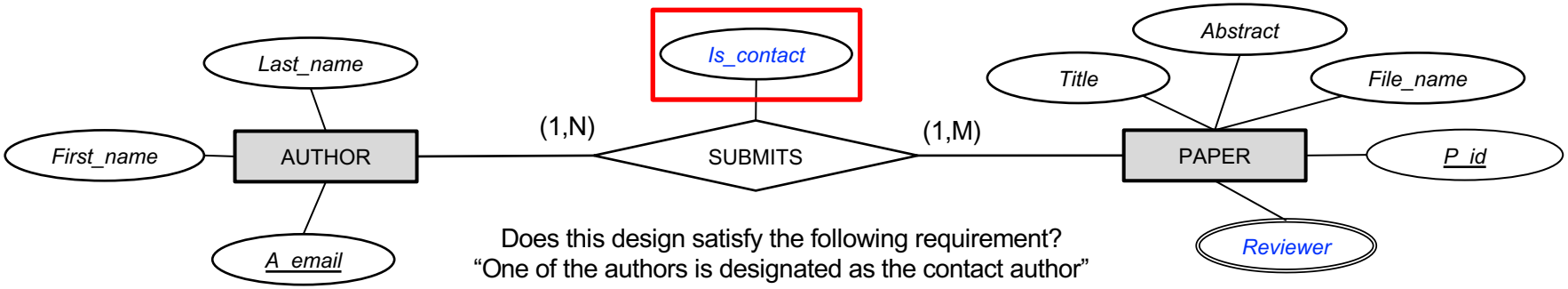
Exercise 2: ER Schema Diagram (Cont.)

Step 2: Describing the Relationship Types

- List and describe relationship types and their cardinality ratios

Relationship Type 1: SUBMITS

- Relating **AUTHOR** and **PAPER** entity types
  - An author is required to submit a paper and a paper has to be submitted by an author to be recorded
    - i.e., both AUTHOR and PAPER types have a total participation (the minimum cardinality = 1)
  - Author can submit many papers and a paper can be submitted by multiple authors (N:M)



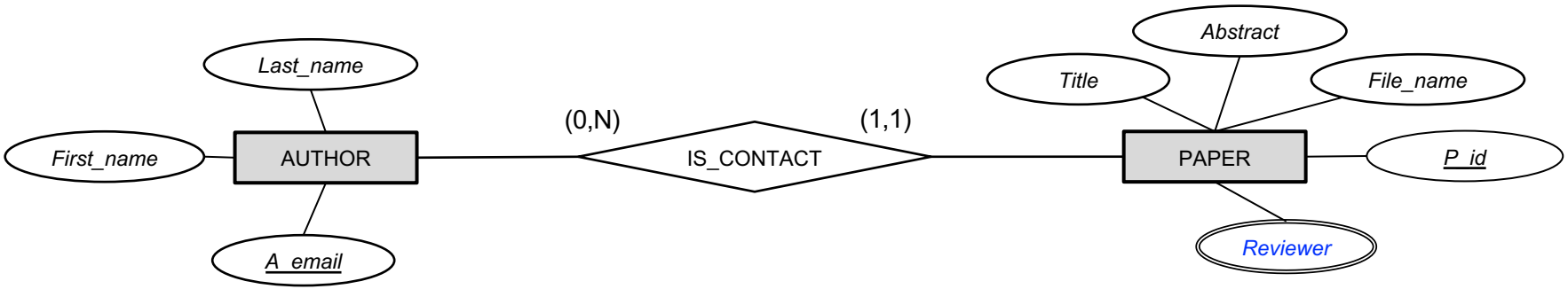
Exercise 2: ER Schema Diagram (Cont.)

Step 2: Describing the Relationship Types

- List and describe relationship types and their cardinality ratios

Relationship Type 2: IS_CONTACT

- Relating **AUTHOR** and **PAPER** entity types
  - For each paper, one author should play the role of "the contact author"
    - i.e., PAPER has a total participation (the minimum cardinality = 1)
  - An author can be the contact for many papers but does not have to be the contact for any papers
    - i.e., AUTHOR has a partial participation (the minimum cardinality = 0)



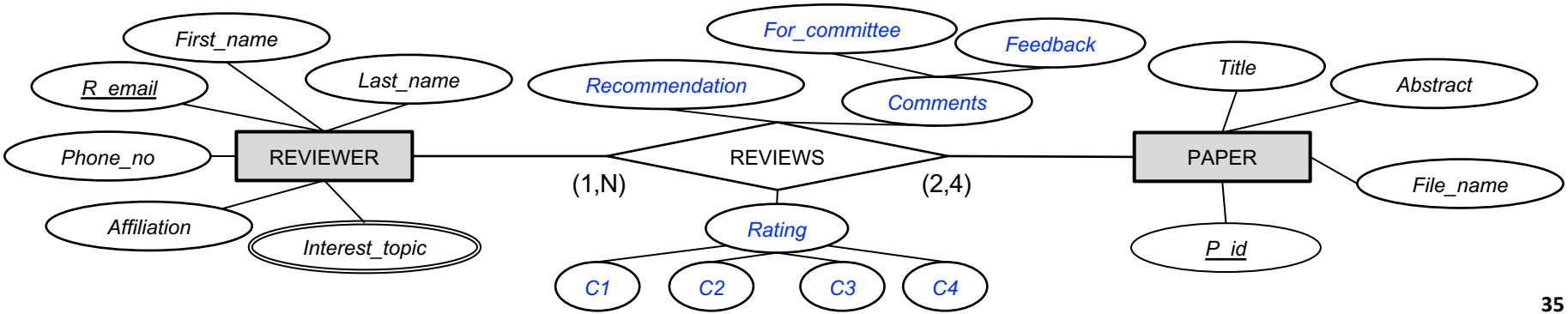
Exercise 2: ER Schema Diagram (Cont.)

Step 2: Describing the Relationship Types

- List and describe relationship types and their cardinality ratios

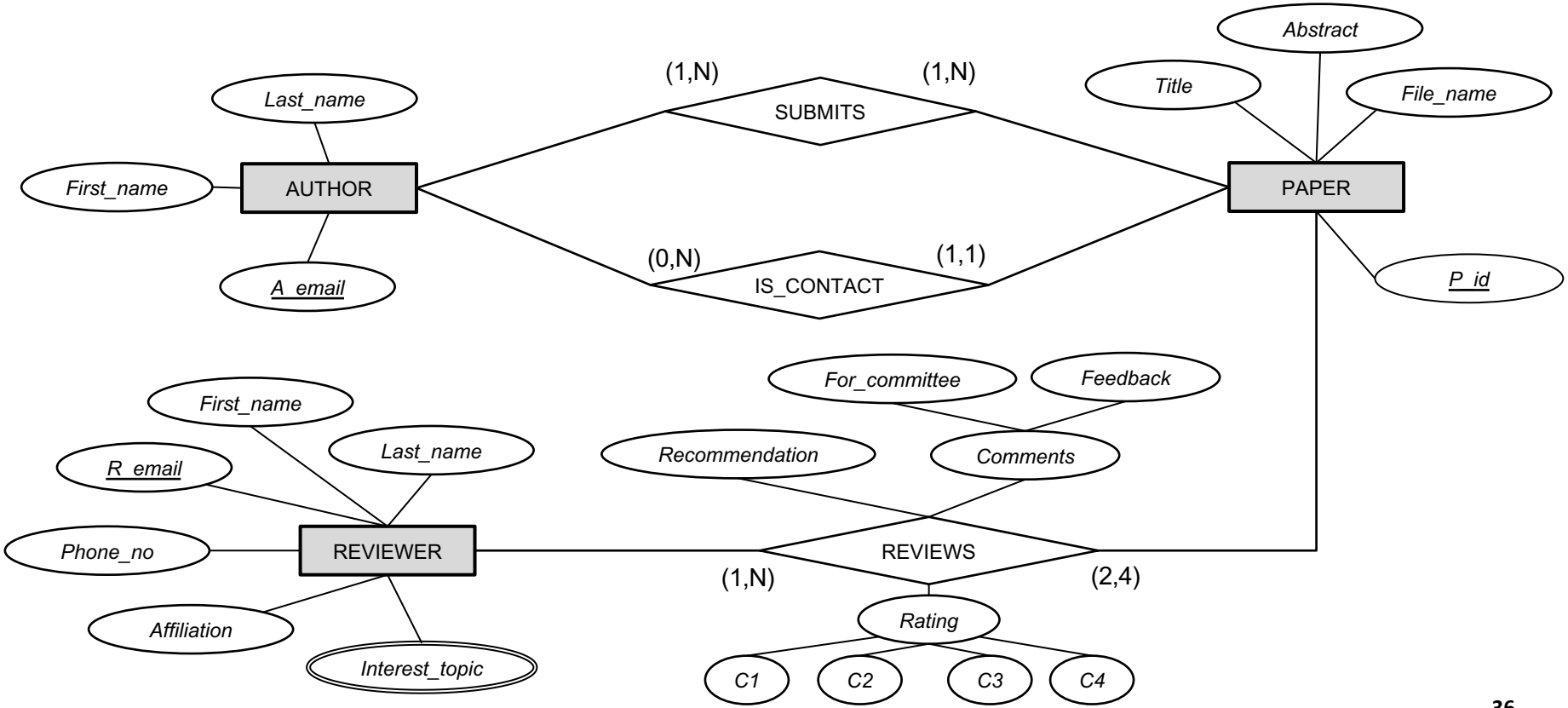
Relationship Type 3: REVIEWS

- Relating **REVIEWER** and **PAPER** entity types for keeping record of reviews
  - It has three relationship attributes including two composite attributes (*Comments* and *Rating*)
  - A paper is reviewed by 2 to 4 reviewers, so its (min, max) constraint is (2,4) (i.e., total participation)
  - Every reviewer has to review at least one paper to be recorded and can review many papers
    - i.e., its (min, max) constraint is (1,N) (i.e., total participation)



Exercise 2: ER Schema Diagram (Cont.)

❑ Step 3: Drawing the ER Diagram



Week 3. Relational Data Model

Exercises: Referential Integrity Constraints

❑ Exercise 1

- Consider the following relations for a database that keeps track of business trips of salespersons in a sales office:

SALESPERSON(Ssn, Name, Start_year, Dept_no)
TRIP(Ssn, From_city, To_city, Departure_date, Return_date, Trip_id)
EXPENSE(Trip_id, Account#, Amount)



- A trip can be charged to one or more accounts
- Specify all foreign keys for this schema

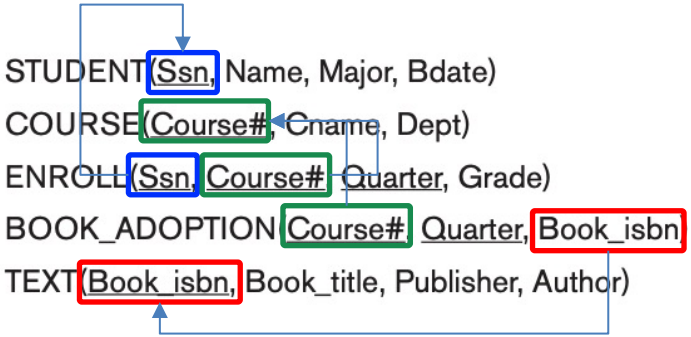
Ssn of the TRIP relation references the SALESPERSON relation:
each trip has to be assigned to an existing salesperson

Trip_id of the EXPENSE relation references the TRIP relation:
An expense has to be paid for a particular trip

Exercises: Referential Integrity Constraints

❑ Exercise 2

- Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:





- Specify all foreign keys for this schema

Ssn of the ENROLL relation references the STUDENT relation:
Each enrollment has to include an existing student

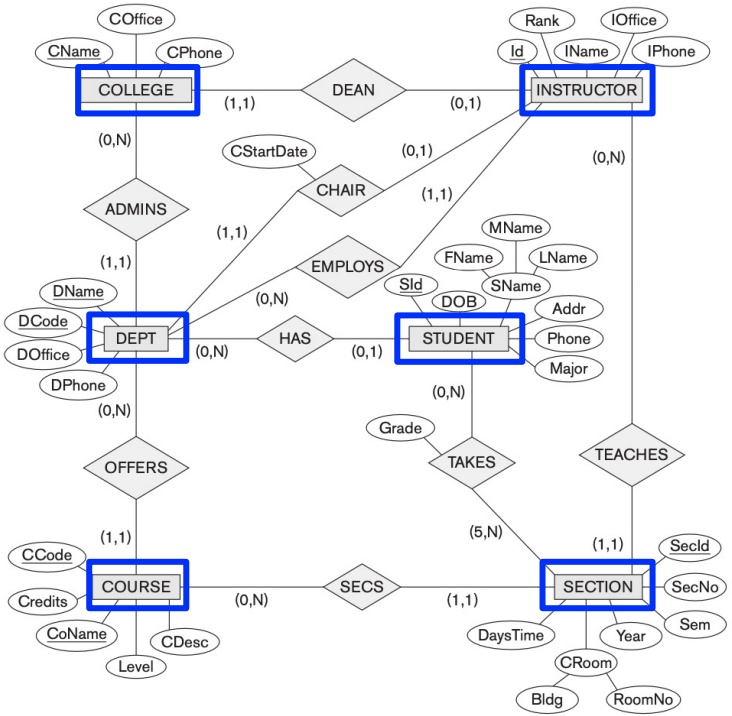
Course# of the ENROLL relation references the COURSE relation:
Each enrollment has to include an existing course

Course# of the BOOK_ADOPTION relation references the COURSE relation:
Each book that is being adopted has to belong to a course

Book_isbn of the BOOK_ADOPTION relation references the TEXT relation:
Each book that is being adopted has to have a valid isbn number that can be found

Exercises: ER-to-Relational Mapping (Cont.)

❑ Step 1: Mapping of Regular Entity Types



COLLEGE

CName	COffice	CPhone

INSTRUCTOR

Id	Rank	IName	IOffice	IPhone

DEPARTMENT

DName	DCode	DOffice	DPhone

COURSE

CCode	Credits	CoName	Level	CDesc

SECTION

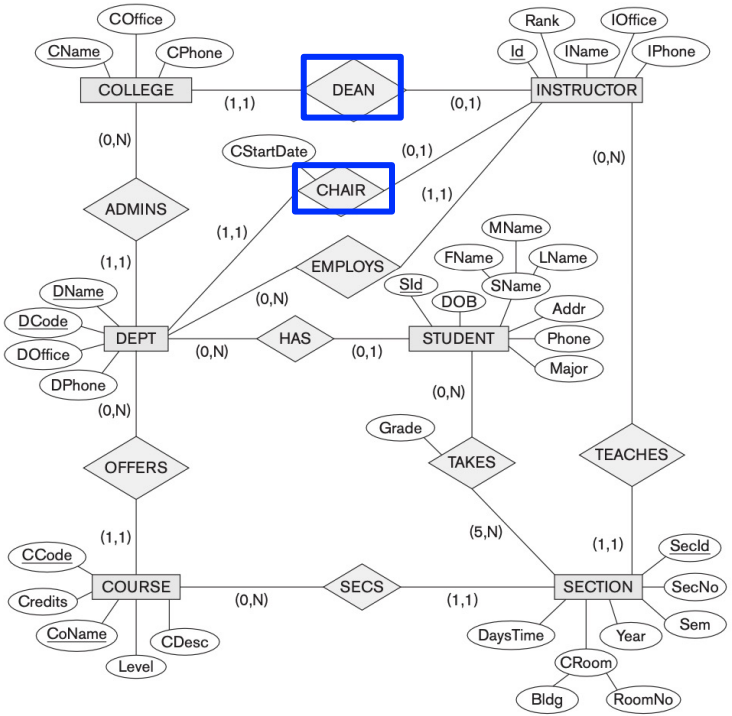
SecId	SecNo	Sem	Year	Bldg	RoomNo	DaysTime

STUDENT

SId	DOB	FName	MName	LName	Addr	Phone	Major

Exercises: ER-to-Relational Mapping (Cont.)

❑ Step 2: Mapping of Binary Relationship Types (1:1)



COLLEGE

CName	COffice	CPhone	DeanId

INSTRUCTOR

Id	Rank	IName	IOffice	IPhone

DEPARTMENT

DName	DCode	DOffice	DPhone	ChairId	CStartDate

COURSE

CCode	Credits	CoName	Level	CDesc

SECTION

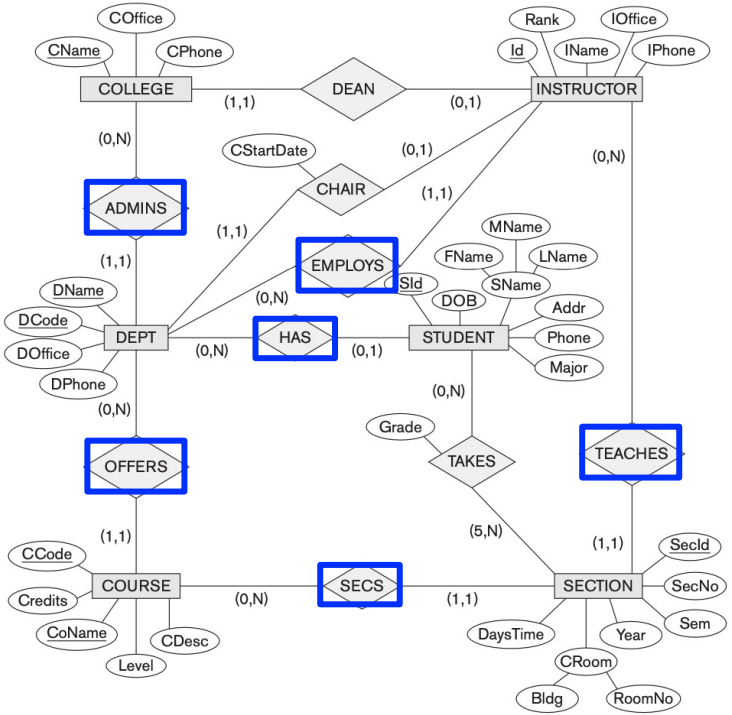
SecId	SecNo	Sem	Year	Bldg	RoomNo	DaysTime

STUDENT

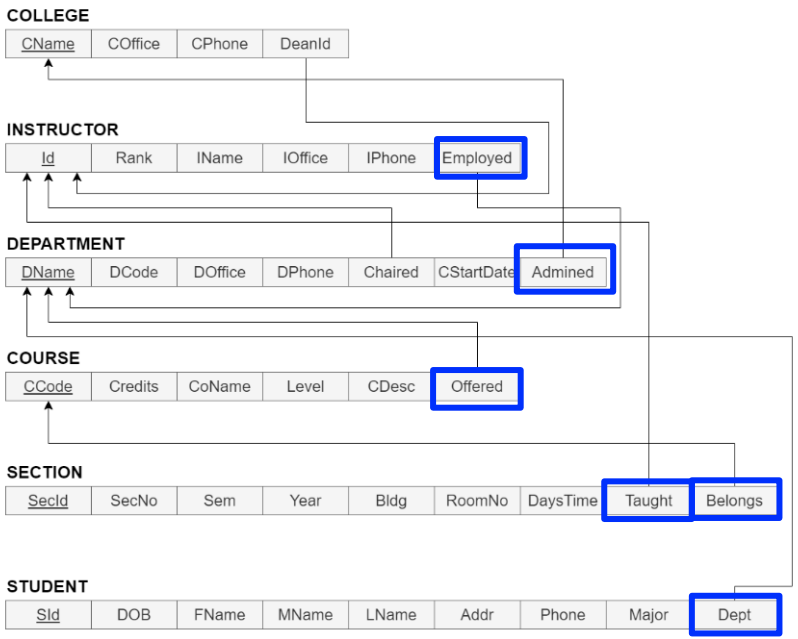
SId	DOB	FName	MName	LName	Addr	Phone	Major

Exercises: ER-to-Relational Mapping (Cont.)

❑ Step 3: Mapping of Binary Relationship Types (1:N)

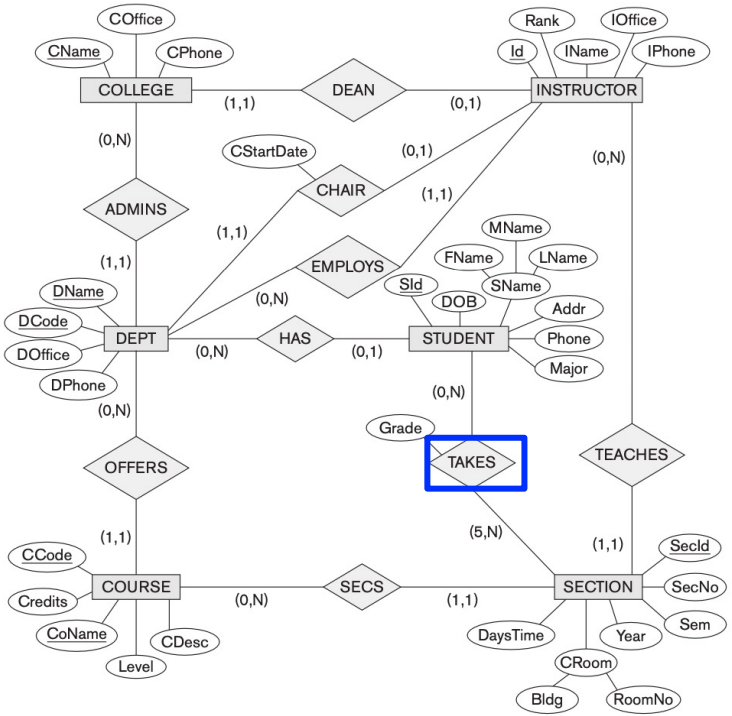






Exercises: ER-to-Relational Mapping (Cont.)

❑ Step 4: Mapping of Binary Relationship Types (M:N)



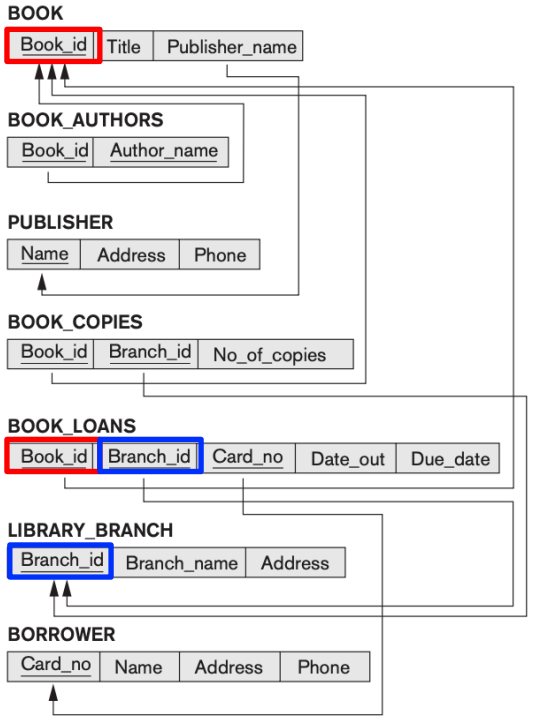


Week 4. Basic SQL

Exercises: Specifying Integrity Constraints

❑ Exercise 1

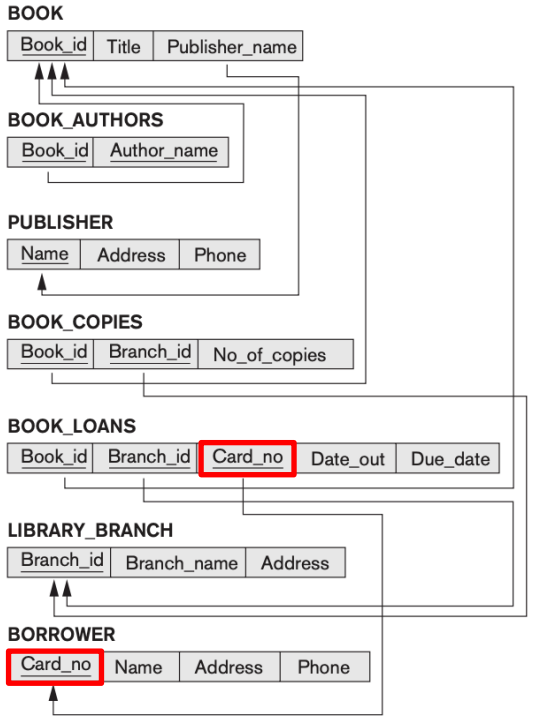
- BOOK_LOANS.Book_id → BOOK.Book_id
  - ❑ **ON DELETE** REJECT (or CASCADE)
    - Imagine if a library administrator deletes a BOOK tuple since the book is removed from the list of all the books in the library
    - However, it does not mean that there are no any loans left, and the library want all books to be returned someday
  - ❑ **ON UPDATE** CASCADE
    - If there is a change in Book_id, the change should be reflected into the referencing tuples in order to correctly keep track of all the book loans
- BOOK_LOANS.Branch_id →LIBRARY_BRANCH.Branch_id
  - ❑ **ON DELETE** CASCADE (or REJECT)
    - The deletion of a LIBRARY_BRANCH tuple: actual "closing" of that branch
    - Thus, the loaned books do not need to be returned (forgive all the loans…)
  - ❑ **ON UPDATE** CASCADE
    - If there is a change in Branch_id, the change should be reflected into the referencing tuples in order to correctly keep track of the library branch where the book is loaned



Exercises: Specifying Integrity Constraints

Exercise 1

- BOOK_LOANS.Card_no → BORROWER.Card_no
  - **ON DELETE** REJECT
    - A library policy do not often permit a member (borrower) to leave the library until the member returns all the loaned books (or paying for them)
    - Thus, the deletion of a referenced tuple (BORROWER) is rejected
  - **ON UPDATE** CASCADE
    - If there is a change in BORROWER.Card_no, the change should be reflected into the referencing tuples (book loans) in order to correctly keep track of the certain borrower who has loaned the book



Exercises: Data Retrieval in SQL

❑ Exercise 2: Specify the Following Queries in SQL

- Q. Retrieve the names of all employees in department 5 who work more than 10 hours per week on the "ProductX" project

Projection attributes

```
SELECT E.Fname, E.Minit, E.Lname
FROM EMPLOYEE AS E, PROJECT AS P, WORKS_ON AS W
WHERE E.Dno=5 AND P.Pname='ProductX' AND W.Hours > 10
AND E.Ssn=W.Essn AND E.Dno=P.Dnum;
```

3 Selection and 2 Join conditions

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Exercises: Data Retrieval in SQL

❑ Exercise 3: Specify the Following Queries in SQL

- Q. Retrieve the name and transcript of each senior student (Class=4) majoring in CS. A transcript includes 1) course name, 2) course number, 3) credit hours, 4) semester, 5) year, and 6) grade for each course completed by the student

```
SELECT   S1.Name, C.Course_name, C.Course_number, C.Credit_hours, S2.Semester, S2.Year, G.Grade
FROM     STUDENT AS S1, COURSE AS C, SECTION AS S2, GRADE_REPORT AS G
WHERE    S1.Class = '4' AND S1.Major = 'CS' AND S1.Student_number = G.Student_number
         AND G.Section_identifier = S2.Section_identifier. AND S2.Course_number = C.Course_number;
```

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Week 5. Advanced SQL

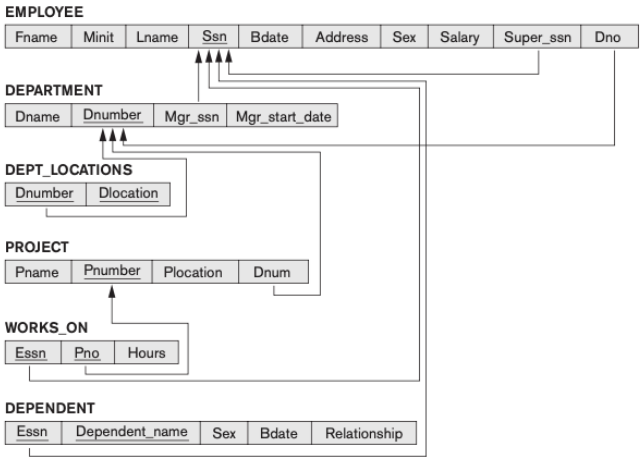
Exercise: Advanced DML

❑ Exercise 1

- Specify the following queries on the COMPANY database using the concept of nested queries, grouping, aggregate functions, etc.

Q1. Retrieve the names of all employees who work in the department that has the employee with the highest salary among all employees

```
SELECT    Fname, Minit, Lname
FROM      EMPLOYEE
WHERE     Dno = (SELECT    Dno
                 FROM      EMPLOYEE
                 WHERE     Salary = (SELECT    MAX(Salary)
                                     FROM      EMPLOYEE ));
```



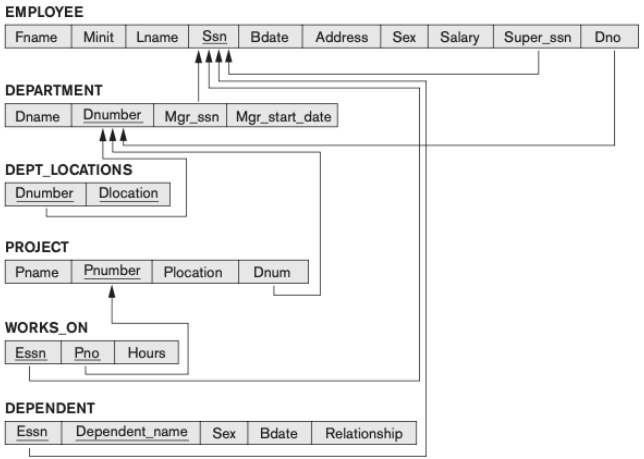
Exercise: Advanced DML

❑ Exercise 1

- Specify the following queries on the COMPANY database using the concept of nested queries, grouping, aggregate functions, etc.

Q2. Retrieve the names of all employees whose supervisor's supervisor has '88866555' for Ssn.

```
SELECT    Fname, Minit, Lname
FROM      EMPLOYEE
WHERE     Super_ssn = ( SELECT    Ssn
                        FROM      EMPLOYEE
                        WHERE     Super_ssn = '88866555' );
```



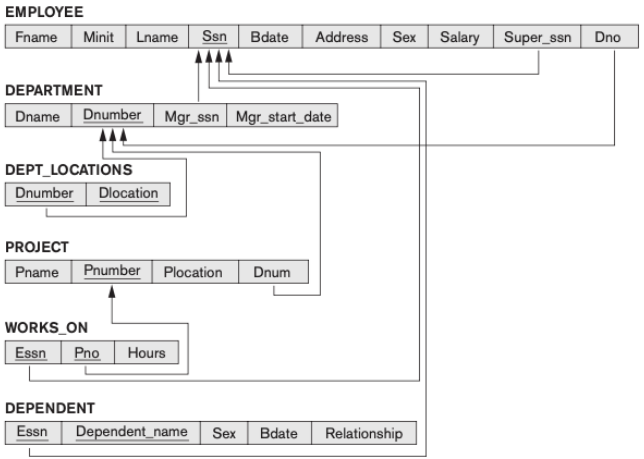
Exercise: Advanced DML

❑ Exercise 1

- Specify the following queries on the COMPANY database using the concept of nested queries, grouping, aggregate functions, etc.

Q3. Retrieve the names of employees who make at least $10,000 more than the employee who is paid the least in the company

```
SELECT    Fname, Minit, Lname
FROM      EMPLOYEE
WHERE     Salary - 10000 >= ( SELECT    MIN(Salary)
                              FROM      EMPLOYEE );
```



Exercise: Advanced DML

❑ Exercise 2

- Specify the following query on the UNIVERSITY database schema in SQL

  Q2-1. Retrieve the names and major department of all straight-A students who have a grade of A in any of their courses

  Q2-2. Retrieve the names and major departments of all students who do not have a grade of A in any of their courses

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

```
SELECT    S.Name, S.Major
FROM      STUDENT AS S
WHERE     NOT EXIST (SELECT *
                     FROM  GRADE_REPORT AS G
                     WHERE  G.Student_number = S.Student_number AND G.Grade <> 'A')
```

```
SELECT Name, Major
FROM STUDENT AS S
WHERE NOT EXIST (SELECT *
                 FROM GRADE_REPORT AS G
                 WHERE G.Student_number = S.Student_number AND G.Grade = 'A')
```

Exercises: Boolean Functions

❑ Exercise 1: Specify the Following Query

Q. Retrieve the name of each employee who works on all the projects controlled by department number 5

```
SELECT    Fname, Lname
FROM      EMPLOYEE
WHERE     NOT EXISTS ( ( SELECT    Pnumber
                         FROM      PROJECT
                         WHERE     Dnum = 5)
                       EXCEPT
                       ( SELECT    Pno
                         FROM      WORKS_ON
                         WHERE     Ssn = Essn) );
```

Set difference operation:
Pnumber and **Pno** are *compatible*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

Exercises: Aggregate Functions

❑ Exercise 2: Specify the following query in SQL

Q. For each department whose average employee salary is more than $30,000, retrieve the department name and the number of employees working for that department

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Fname	Minit	Lname	Ssn	...	Salary	Super_ssn	Dno
John	B	Smith	123456789		30000	333445555	5
Franklin	T	Wong	333445555		40000	888665555	5
Ramesh	K	Narayan	666884444		38000	333445555	5
Joyce	A	English	453453453	...	25000	333445555	5
Alicia	J	Zelaya	999887777		25000	987654321	4
Jennifer	S	Wallace	987654321		43000	888665555	4
Ahmad	V	Jabbar	987987987		25000	987654321	4
James	E	Bong	888665555		55000	NULL	1

```
SELECT    D.Dname, COUNT(*)
FROM      EMPLOYEE AS E, DEPARTMENT AS D
WHERE     E.Dno = D.Dnumber
GROUP BY  E.Dno
HAVING    AVG(E.Salary) > 30000
```

Exercises: Views in SQL

❑ Exercise 3

A view that has the department name, manager name, and manager salary for every department

→ A view for describing each department and its manager

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

```
CREATE VIEW DEPT_INFO(DeptName, MgrFname, MgrLname, MgrSalary)
AS  SELECT  D.Dname, E.Fname, E.Lname, E.Salary
    FROM    DEPARTMENT AS D, EMPLOYEE AS E
    WHERE   D.Mgr_ssn=E.Ssn
```

DEPT_INFO

Research	Franklin	T	Wong	40000
Administration	Jennifer	S	Wallace	43000
Headquarters	James	E	Borg	55000

Exercises: Views in SQL

❑ Exercise 4

A view that has the project name, controlling department name, number of employees, and total hours worked per week on the project for each project

→ A view for describing each project and its current status

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

```
CREATE VIEW ProjectInfo(Pname, Dname, NumEmployees, TotalHours)
AS  SELECT   P.Pname, D.Dname, COUNT(*), SUM(W.Hours)
    FROM     PROJECT AS P, WORKS_ON AS W, DEPARTMENT AS D
    WHERE    P.Pnumber = W.Pno AND D.Dnumber = P.Dnum
    GROUP BY P.Pnumber, D.Dname
```

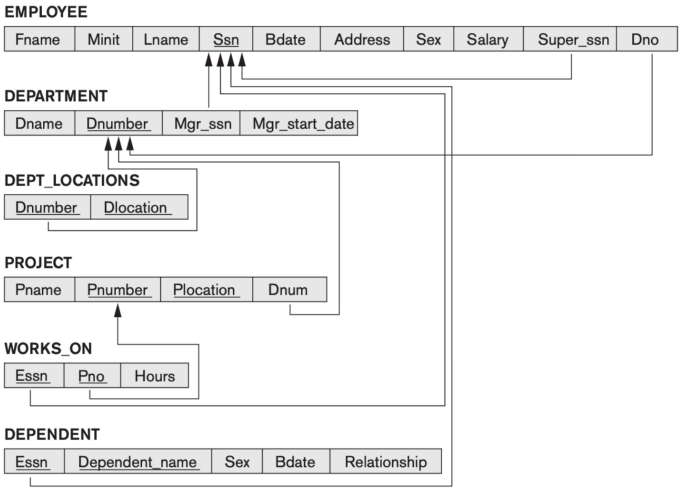
Compute the number of employees and total hours
Per each group

Week 7. Relational Algebra

Exercises: Relational Algebra

❑ Exercise 1: State the Following Query in Relational Algebra

- Q. Retrieve the name and address of all employees who work for the 'Research' department

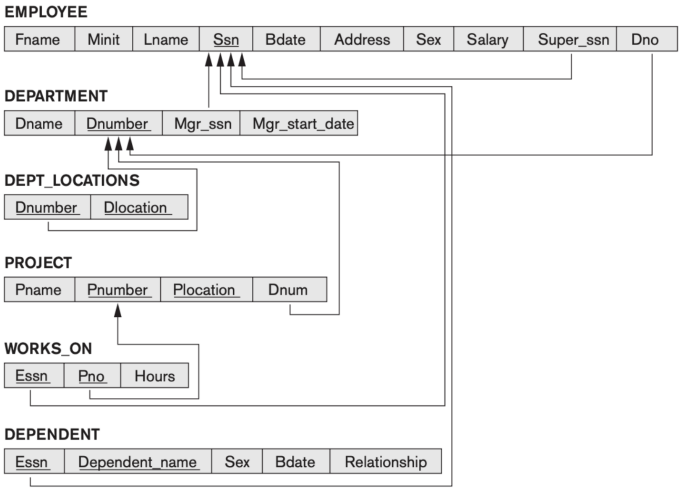


$$\text{RESEARCH_DEPT} \leftarrow \sigma_{\text{Dname='Research'}}(\text{DEPARTMENT})$$
$$\text{RESEARCH_EMPS} \leftarrow (\text{RESEARCH_DEPT} \bowtie_{\text{Dnumber=Dno}} \text{EMPLOYEE})$$
$$\text{RESULT} \leftarrow \pi_{\text{Fname, Lname, Address}}(\text{RESEARCH_EMPS})$$

Exercises: Relational Algebra

❑ Exercise 2: State the Following Query in Relational Algebra

- Q. For every project located in “Stafford”, list the project number, the controlling department number, and the department manager’s last name, address, and birth date

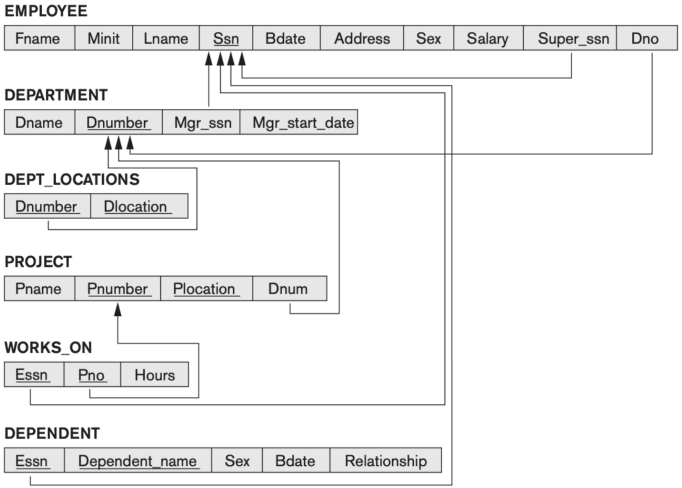


$$\text{STAFFORD_PROJS} \leftarrow \sigma_{\text{Plocation='Stafford'}}(\text{PROJECT})$$
$$\text{CONTR_DEPTS} \leftarrow (\text{STAFFORD_PROJS} \bowtie_{\text{Dnum=Dnumber}} \text{DEPARTMENT})$$
$$\text{PROJ_DEPT_MGRS} \leftarrow (\text{CONTR_DEPTS} \bowtie_{\text{Mgr_ssn=SsnE}} \text{MPLOYEE})$$
$$\text{RESULT} \leftarrow \pi_{\text{Pnumber, Dnum, Lname, Address, Bdate}}(\text{PROJ_DEPT_MGRS})$$

Exercises: Relational Algebra

❑ Exercise 3: State the Following Query in Relational Algebra

- Q. Make a list of project numbers for projects that involve an employee whose last name is ‘Smith’, either as a worker or as a manager of the department that controls the project



Get the project numbers as a worker

$SMITHS(Essn) \leftarrow \pi_{Ssn}(\sigma_{Lname='Smith'}(EMPLOYEE))$

$SMITH_WORKER_PROJS \leftarrow \pi_{Pno}(WORKS_ON * SMITHS)$

Get the project numbers as a manager

$MGRS \leftarrow \pi_{Lname,\ Dnumber}(EMPLOYEE \bowtie_{Ssn=Mgr_ssn} DEPARTMENT)$

$SMITH_MANAGED_DEPTS(Dnum) \leftarrow \pi_{Dnumber}(\sigma_{Lname='Smith'}(MGRS))$

$SMITH_MGR_PROJS(Pno) \leftarrow \pi_{Pnumber}(SMITH_MANAGED_DEPTS * PROJECT)$

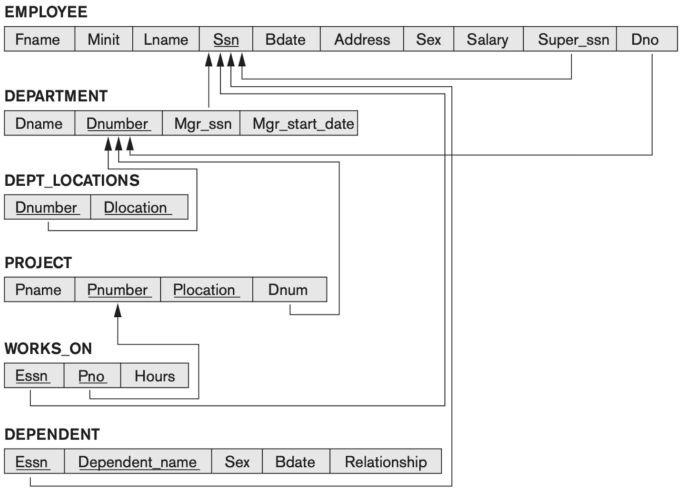
Combine the results

$RESULT \leftarrow (SMITH_WORKER_PROJS \cup SMITH_MGR_PROJS)$

Exercises: Relational Algebra

❑ Exercise 4: State the Following Query in Relational Algebra

- Q. Retrieve the names of employees who have no dependents

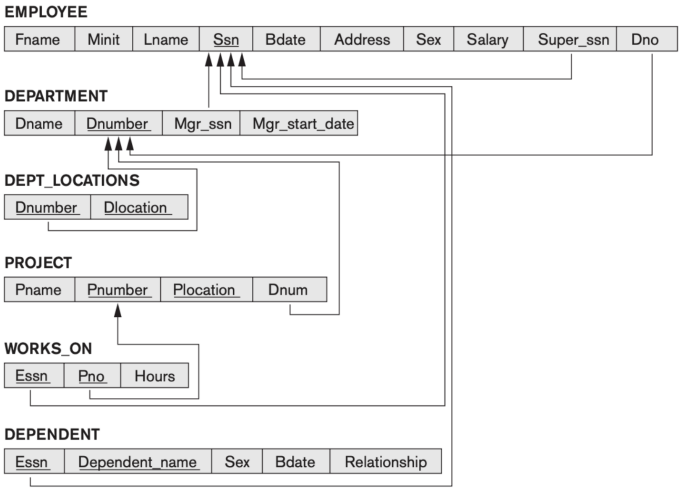


$$ALL_EMPS \leftarrow \pi_{Ssn}(EMPLOYEE)$$
$$EMPS_WITH_DEPS(Ssn) \leftarrow \pi_{Essn}(DEPENDENT)$$
$$EMPS_WITHOUT_DEPS \leftarrow (ALL_EMPS - EMPS_WITH_DEPS)$$
$$RESULT \leftarrow \pi_{Lname,\ Fname}(EMPS_WITHOUT_DEPS * EMPLOYEE)$$

Exercises: Relational Algebra

❑ Exercise 5: State the Following Query in Relational Algebra

- Q. List the names of managers who have at least one dependent



$$MGRS(Ssn) \leftarrow \pi_{Mgr_ssn}(DEPARTMENT)$$
$$EMPS_WITH_DEPS(Ssn) \leftarrow \pi_{Essn}(DEPENDENT)$$
$$MGRS_WITH_DEPS \leftarrow (MGRS \cap EMPS_WITH_DEPS)$$
$$RESULT \leftarrow \pi_{Lname,\ Fname}(MGRS_WITH_DEPS * EMPLOYEE)$$